

Differential Equations

Lecture Notes & Foundations

Differential Equations (DE):

An equation involving derivatives of one or more dependent variables with respect to one or more independent variables is called a **differential equation**.

Ordinary Differential Equation (ODE):

A differential equation involving ordinary derivatives of one or more dependent variables with respect to a single independent variable is called an **ordinary differential equation**.

Partial Differential Equation (PDE):

A differential equation involving partial derivatives of one or more dependent variables with respect to more than one independent variable is called a **partial differential equation**.

EXAMPLES OF DIFFERENTIAL EQUATIONS

Ordinary Differential Equations (ODE)

$$\frac{d^2y}{dx^2} + xy \left(\frac{dy}{dx} \right)^2 = 0$$

$$\frac{d^4x}{dt^4} + 5\frac{d^2x}{dt^2} + 3x = \sin(t)$$

Partial Differential Equations (PDE)

$$\frac{\partial v}{\partial s} + \frac{\partial v}{\partial t} = v$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

Basic Derivatives (Not Differential Equations)

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

Order of a DE:

The order of the highest-ordered derivative involved in a differential equation is called the **order** of the differential equation.

Degree of a DE:

The power (or degree) of the highest-ordered derivative involved in a differential equation is called the **degree** of the differential equation.

CLASSIFICATION BY ORDER AND DEGREE

2nd Order, 1st Degree ODE

$$\frac{d^2y}{dx^2} + xy \left(\frac{dy}{dx} \right)^2 = 0$$

4th Order, 1st Degree ODE

$$\frac{d^4x}{dt^4} + 5 \frac{d^2x}{dt^2} + 3x = \sin(t)$$

1st Order, 1st Degree PDE

$$\frac{\partial v}{\partial s} + \frac{\partial v}{\partial t} = v$$

2nd Order, 1st Degree PDE

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

2nd Order, 2nd Degree PDE

$$\left(\frac{\partial^2 z}{\partial x^2} \right)^2 + \left(\frac{\partial^2 z}{\partial y^2} \right)^2 = 0$$

Linear ODE:

A linear ordinary differential equation of order n , in the dependent variable y and the independent variable x , is an equation that can be expressed in the form:

$$a_0(x)\frac{d^n y}{dx^n} + a_1(x)\frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_{n-1}(x)\frac{dy}{dx} + a_n(x)y = b(x)$$

where $a_0(x)$ is not identically zero.

Key Observations for Linearity:

- 1) The dependent variable y and its various derivatives occur to the 1st degree only.
- 2) No products of y and/or any of its derivatives are present.
- 3) No transcendental functions of y or its derivatives occur (e.g., $\sin y$, e^y).

Nonlinear ODE:

A non-linear ordinary differential equation is an ordinary differential equation that is not linear.

EXAMPLES OF LINEAR DIFFERENTIAL EQUATIONS

(a) Linear ODE with Constant Coefficients

$$\frac{d^2 y}{dx^2} + 5\frac{dy}{dx} + 6y = 0 \quad \text{Coefficients: } (1, 5, 6)$$

(b) Linear ODE with Variable Coefficients

$$\frac{d^4 y}{dx^4} + x^2 \frac{d^3 y}{dx^3} + x^3 \frac{dy}{dx} = xe^x \quad \text{Coefficients: } (1, x^2, x^3)$$

Examples of Nonlinear ODEs:

1)

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y^2 = 0$$

Reason: The dependent variable y is raised to the second power (y^2), violating linearity.

2)

$$\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 + 6y = 0$$

Reason: The derivative $\frac{dy}{dx}$ is raised to the third power, violating linearity.

3)

$$\frac{d^2y}{dx^2} + 5y\frac{dy}{dx} + 6y = 0$$

Reason: The dependent variable y and its derivative $\frac{dy}{dx}$ are multiplied together.

Origin and Applications:

Differential equations occur in connection with numerous problems encountered across various branches of science and engineering:

- Population Dynamics
- Dynamical Systems and Chaos
- Fluid Dynamics
- Meteorology
- Heat Transfer
- Biophysics
- ...etc.

Formation of Differential Equations:

Example Problem:

Form the differential equation from $y = ax + bx^2$ by eliminating the arbitrary constants a and b .

Solution:

Given the relation:

$$y = ax + bx^2 \tag{1}$$

Differentiating with respect to x :

$$\frac{dy}{dx} = a + 2bx \implies a = \frac{dy}{dx} - 2bx \tag{2}$$

Differentiating again with respect to x :

$$\frac{d^2y}{dx^2} = 2b \implies b = \frac{1}{2} \frac{d^2y}{dx^2} \tag{3}$$

Substituting (3) into (2):

$$a = \frac{dy}{dx} - x \frac{d^2y}{dx^2}$$

Substituting a and b back into (1):

$$y = x \left(\frac{dy}{dx} - x \frac{d^2y}{dx^2} \right) + \frac{1}{2} x^2 \frac{d^2y}{dx^2} = x \frac{dy}{dx} - \frac{1}{2} x^2 \frac{d^2y}{dx^2}$$

Rearranging terms yields the final 2nd order ODE:

$$x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$$

General Solution:

The relation containing n arbitrary constants which satisfies an ODE of n^{th} order is called its **complete primitive** or **general solution**.

Note: It can be shown that, by eliminating n arbitrary constants from an equation in x and y , we obtain a differential equation of n^{th} order. Such a process is called the **formation of a differential equation**.

Particular Solution:

A **particular solution** is one obtained from the primitive or general solution by assigning definite values to the arbitrary constants.

Geometrical Interpretation:

Geometrically, the primitive represents the equation of a **family of curves** satisfying the differential equation, and a particular solution represents the equation of **one member** of this family of curves.

Example Problem:

Form the differential equation from the family of parabolas:

$$(x - h)^2 = 4a(y - k) \quad (4)$$

where a, h, k are arbitrary constants.

Solution:

Differentiating (4) with respect to x :

$$2(x - h) = 4a \frac{dy}{dx} \implies (x - h) = 2a \frac{dy}{dx} \quad (5)$$

Differentiating (5) again with respect to x :

$$1 = 2a \frac{d^2y}{dx^2} \implies \frac{d^2y}{dx^2} = \frac{1}{2a} \quad (6)$$

Differentiating (6) once more with respect to x :

$$\frac{d^3y}{dx^3} = 0$$

which is the required differential equation of order 3.

Nature of Solutions:

Consider the general n^{th} order differential equation:

$$F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}\right) = 0 \quad (7)$$

where F is a real-valued function of its $(n + 2)$ arguments.

- 1. Explicit Solution:** Let f be a real function defined for all x in an interval I and having an n^{th} derivative for all $x \in I$. The function f is called an **explicit solution** of the DE (7) on I if it fulfills the following two requirements:

- i) $F(x, f(x), f'(x), \dots, f^{(n)}(x))$ is defined for all $x \in I$.
- ii) $F(x, f(x), f'(x), \dots, f^{(n)}(x)) = 0 \quad \forall x \in I$.

- 2. Implicit Solution:** A relation $g(x, y) = 0$ is called an **implicit solution** of the DE (7) if this relation defines at least one real function f of variable x on I such that this function is an explicit solution of (7) on I .
- 3.** Both explicit and implicit solutions will usually be called simply **solutions**.

Example 1:

Show that $f(x) = 2 \sin x + 3 \cos x$ is an explicit solution of $\frac{d^2 y}{dx^2} + y = 0$ for all x .

Solution:

Given $f(x) = 2 \sin x + 3 \cos x$. The derivatives are:

$$f'(x) = 2 \cos x - 3 \sin x, \quad f''(x) = -2 \sin x - 3 \cos x$$

Substituting into the DE:

$$\frac{d^2 y}{dx^2} + y = (-2 \sin x - 3 \cos x) + (2 \sin x + 3 \cos x) = 0$$

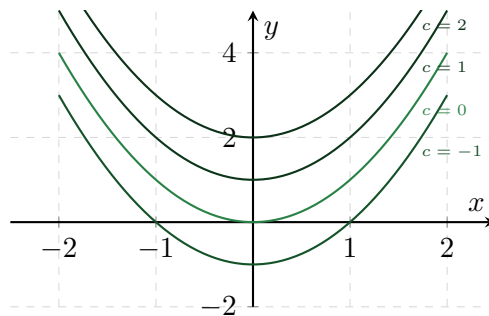
This holds for all real x . Therefore, $f(x)$ is an explicit solution.

Example 2:

Consider $\frac{dy}{dx} = 2x$. This differential equation has a one-parameter family of solutions:

$$y = x^2 + c$$

where c is an arbitrary constant (parameter).



Exercise:

Show that $f(x) = c_1 e^{4x} + c_2 e^{-2x}$ is a solution of $y'' - 2y' - 8y = 0$.

Solution:

$$f'(x) = 4c_1 e^{4x} - 2c_2 e^{-2x} \text{ and } f''(x) = 16c_1 e^{4x} + 4c_2 e^{-2x}.$$

$$y'' - 2y' - 8y = (16c_1 e^{4x} + 4c_2 e^{-2x}) - 2(4c_1 e^{4x} - 2c_2 e^{-2x}) - 8(c_1 e^{4x} + c_2 e^{-2x}) = 0$$

Thus, $f(x)$ is an explicit solution for all real x .

Initial Value Problem (IVP) & Boundary Value Problem (BVP)

Example:

Consider $\frac{dy}{dx} = 2x$ — (1) subject to $y(1) = 4$ — (2).

Solution:

Equation (1) has a 1-parameter family of solutions:

$$y = x^2 + c \quad (8)$$

Applying condition (2) by substituting $x = 1$:

$$y(1) = 1^2 + c \implies 4 = 1 + c \implies c = 3$$

Thus, $y = x^2 + 3$ is the unique solution of (1) satisfying condition (2).

Supplementary Conditions:

In practical applications of both first and higher-order differential equations, problems involve a DE along with one or more supplementary conditions which the solution must satisfy.

Initial-Value Problem (IVP):

If all associated supplementary conditions relate to **one** single value of x , the problem is called an **initial-value problem** (or one-point boundary value problem).

Boundary Value Problem (BVP):

If the supplementary conditions relate to **two different** values of x , the problem is called a **two-point boundary value problem** or simply a **boundary value problem**.

Comparison Examples:

Initial Value Problem (IVP):

$$\frac{d^2y}{dx^2} + y = 0, \quad y(1) = 3, \quad y'(1) = 4 \quad (\text{Conditions evaluated at single point } x = 1)$$

Boundary Value Problem (BVP):

$$\frac{d^2y}{dx^2} + y = 0, \quad y(0) = 1, \quad y\left(\frac{\pi}{2}\right) = 5 \quad (\text{Conditions evaluated at } x = 0 \text{ and } x = \frac{\pi}{2})$$

Existence and Uniqueness Theorem

Consider the first-order differential equation:

$$\frac{dy}{dx} = f(x, y) \quad \text{--- } (*)$$

where:

- 1) $f(x, y)$ is a continuous function in some domain D of the xy -plane, and
- 2) $\frac{\partial f}{\partial y}$ is also a continuous function of x and y in D .

Let (x_0, y_0) be an interior point of D . Then there exists a **unique solution** $\phi(x)$ of the DE $(*)$ defined on some interval $|x - x_0| \leq h$ (where $h > 0$ is sufficiently small) that satisfies the initial condition:

$$\phi(x_0) = y_0$$

Reference: S.L. Ross, Differential Equations, 3rd Edition, p. 21.

Formation of Differential Equations

Method: Eliminating ordinary arbitrary constants.

Example 1:

Form the differential equation by eliminating arbitrary constants a and b from:

$$y = ax + bx^2$$

Solution:

Given $y = ax + bx^2$ — (1). Differentiating once: $\frac{dy}{dx} = a + 2bx$ — (2).

Differentiating again: $\frac{d^2y}{dx^2} = 2b \implies b = \frac{1}{2} \frac{d^2y}{dx^2}$ — (3).

From (2): $a = \frac{dy}{dx} - 2bx = \frac{dy}{dx} - x \frac{d^2y}{dx^2}$ — (4).

Substituting (3) and (4) into (1):

$$y = x \left(\frac{dy}{dx} - x \frac{d^2y}{dx^2} \right) + \frac{1}{2} x^2 \frac{d^2y}{dx^2} \implies x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$$

Example 2:

Form the differential equation of all parabolas whose axes are parallel to the y -axis.

Solution:

The general equation of parabolas with vertical axes is given by:

$$(x - h)^2 = 4a(y - k) \tag{9}$$

where a, h, k are three arbitrary constants.

Differentiating (9) with respect to x :

$$2(x - h) = 4a \frac{dy}{dx} \implies (x - h) = 2a \frac{dy}{dx}$$

Differentiating a second time:

$$1 = 2a \frac{d^2y}{dx^2} \implies \frac{d^2y}{dx^2} = \frac{1}{2a}$$

Differentiating a third time:

$$\frac{d^3y}{dx^3} = 0$$

which is the required differential equation of 3rd order.

First Order & First Degree Differential Equations

Lecture Notes — Section 1: Separable Equations

Equations of First Order and 1st Degree:

The 1st order differential equation can be expressed as:

$$\frac{dy}{dx} = f(x, y) \quad \text{— (i) (Derivative Form)} \quad (10)$$

$$\text{or, } M(x, y)dx + N(x, y)dy = 0 \quad \text{— (ii) (Differential Form)} \quad (11)$$

An equation in one of these forms may readily be written in the other form.

Separable Equations and Equations Reducible to This Form:

An equation of the form

$$F(x)G(y)dx + f(x)g(y)dy = 0$$

is called an **equation with variables separable** or simply a **separable equation**.

Example 1: Solve $\sec^2 x \tan y \, dx + \sec^2 y \tan x \, dy = 0$

Solution: Separating the variables, we get:

$$\frac{\sec^2 x}{\tan x} dx + \frac{\sec^2 y}{\tan y} dy = 0$$

Integrating both sides:

$$\log(\tan x) + \log(\tan y) = c_0 \implies \log(\tan y \tan x) = c_0$$

$$\implies \tan x \tan y = e^{c_0} = c \quad \text{Ans.}$$

Initial Value Problem (IVP):

Solve the IVP:

$$x \sin y \, dx + (x^2 + 1) \cos y \, dy = 0 \quad \text{— (I)} \quad (12)$$

with initial condition $y(1) = \frac{\pi}{2}$ — (II)

Solution: Separating the variables:

$$\frac{x}{x^2 + 1} dx + \frac{\cos y}{\sin y} dy = 0 \quad [\text{assuming } y \neq 0]$$

Integrating both sides:

$$\frac{1}{2} \ln(x^2 + 1) + \ln(\sin y) = \ln c_0$$

IVP Solution (Continued):

Multiplying by 2:

$$\begin{aligned}\ln(x^2 + 1) + 2\ln(\sin y) &= 2\ln c_0 \implies \ln[(x^2 + 1)\sin^2 y] = \ln c \\ \implies (x^2 + 1)\sin^2 y &= c \quad \text{--- (III)}\end{aligned}$$

By using boundary conditions (II) in (III):

$$(1^2 + 1)\sin^2\left(\frac{\pi}{2}\right) = c \implies 2(1)^2 = c \implies c = 2$$

Substituting $c = 2$ back into (III):

$$(x^2 + 1)\sin^2 y = 2$$

which is the required solution.

Equations Reducible to the Form of Variable Separable:

Example 2: Solve $\frac{dy}{dx} = (4x + y + 1)^2$ --- (1)

Solution: Let $4x + y + 1 = u \implies 4 + \frac{dy}{dx} = \frac{du}{dx} \implies \frac{dy}{dx} = \frac{du}{dx} - 4$.
Substituting into (1):

$$\frac{du}{dx} - 4 = u^2 \implies \frac{du}{dx} = u^2 + 4 \implies \frac{du}{u^2 + 2^2} = dx \quad \text{[Variable Separable]}$$

Integrating both sides:

$$\frac{1}{2} \tan^{-1}\left(\frac{u}{2}\right) = x + c \implies \frac{1}{2} \tan^{-1}\left(\frac{4x + y + 1}{2}\right) = x + c \quad \text{Ans.}$$

Example 3: Solve $(x + y)^2 \frac{dy}{dx} = a^2$ --- (1)

Solution: Let $x + y = u \implies 1 + \frac{dy}{dx} = \frac{du}{dx} \implies \frac{dy}{dx} = \frac{du}{dx} - 1$.
Substituting into (1):

$$\begin{aligned}u^2 \left(\frac{du}{dx} - 1 \right) &= a^2 \implies u^2 \frac{du}{dx} - u^2 = a^2 \implies \frac{du}{dx} = \frac{a^2 + u^2}{u^2} \\ \implies \frac{u^2}{a^2 + u^2} du &= dx \implies \left(1 - \frac{a^2}{a^2 + u^2} \right) du = dx\end{aligned}$$

Integrating both sides:

$$u - a^2 \cdot \frac{1}{a} \tan^{-1}\left(\frac{u}{a}\right) = x + c$$

Reducible Equations (Continued):

Substituting back $u = x + y$:

$$(x + y) - a \tan^{-1} \left(\frac{x + y}{a} \right) = x + c \implies y = a \tan^{-1} \left(\frac{x + y}{a} \right) + c \quad \text{Ans.}$$

Homogeneous Differential Equations:

The 1st order differential equation $M(x, y)dx + N(x, y)dy = 0$ is said to be **homogeneous** if it can be written in the form:

$$\frac{dy}{dx} = \frac{f_1(x, y)}{f_2(x, y)}$$

in which $f_1(x, y)$ and $f_2(x, y)$ are homogeneous functions of the same degree.

The homogeneous equation can be reduced to an equation in which variables are separable by substitution:

$$y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Alternative Test for Homogeneity:

If $\frac{dy}{dx} = f(x, y)$, then it is homogeneous if $f(x, y)$ can be expressed in the form $g\left(\frac{y}{x}\right)$.

Example 1 (Homogeneous):

$$\text{Solve } (x^2 + y^2)dx + 2xy dy = 0 \quad \text{--- (1)}$$

Solution: Rewriting (1):

$$(1) \implies \frac{dy}{dx} = -\frac{x^2 + y^2}{2xy} \quad (\text{Homogeneous}) \longrightarrow g\left(\frac{y}{x}\right)$$

$$\left[\text{Check degree: } f(kx, ky) = -\frac{(kx)^2 + (ky)^2}{2(kx)(ky)} = -\frac{x^2 + y^2}{2xy} \implies n = 0 \text{ (Zero Degree)} \right]$$

Homogeneous DE Solutions (Continued):

Substituting $y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx}$ — (2)
Substituting (2) into (1):

$$\begin{aligned} v + x \frac{dv}{dx} &= -\frac{x^2 + x^2 v^2}{2x \cdot vx} = -\frac{1 + v^2}{2v} \\ \implies x \frac{dv}{dx} &= -\frac{1 + v^2}{2v} - v = \frac{-1 - v^2 - 2v^2}{2v} = -\frac{1 + 3v^2}{2v} \\ \implies \frac{dx}{x} &= -\frac{2v}{1 + 3v^2} dv \quad [\text{Variable Separable}] \end{aligned}$$

Integrating both sides:

$$\begin{aligned} \ln x &= -\frac{1}{3} \ln(1 + 3v^2) + \ln c \implies \ln x + \ln(1 + 3v^2)^{1/3} = \ln c \\ \implies x(1 + 3v^2)^{1/3} &= c \implies x \left(1 + 3\frac{y^2}{x^2}\right)^{1/3} = c \quad \mathbf{Ans.} \end{aligned}$$

Example 2 (Homogeneous):

Solve $(x^2 - 3y^2)dx + 2xy dy = 0$

Solution:

$$\frac{dy}{dx} = -\frac{x^2 - 3y^2}{2xy} = -\frac{x}{2y} + \frac{3y}{2x} = -\frac{1}{2} \left(\frac{1}{y/x}\right) + \frac{3}{2} \left(\frac{y}{x}\right) \quad \left[\text{Homogeneous in } g\left(\frac{y}{x}\right)\right]$$

Substituting $y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx}$:

$$\begin{aligned} v + x \frac{dv}{dx} &= -\frac{1}{2v} + \frac{3}{2}v \implies x \frac{dv}{dx} = -\frac{1}{2v} + \frac{3}{2}v - v = \frac{v^2 - 1}{2v} \\ \implies \frac{dx}{x} &= \frac{2v dv}{v^2 - 1} \end{aligned}$$

Integrating both sides:

$$\begin{aligned} \ln x &= \ln |v^2 - 1| - \ln c \implies v^2 - 1 = cx \implies \left(\frac{y}{x}\right)^2 - 1 = cx \\ \implies y^2 - x^2 &= cx^3 \quad \mathbf{Ans.} \end{aligned}$$

Equations Reducible to Homogeneous Form:

An equation of the form:

$$\frac{dy}{dx} = \frac{ax + by + c}{a_1x + b_1y + c_1}, \quad \text{where } \frac{a}{a_1} \neq \frac{b}{b_1}$$

can be reduced to homogeneous form by substituting:

$$x = X + h, \quad y = Y + k \implies \frac{dy}{dx} = \frac{dY}{dX}$$

Example: Solve $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$ — (1)

Solution: Put $x = X + h$ and $y = Y + k$, where h, k are constants.

$$\text{Then, } \frac{dy}{dx} = \frac{dY}{dX}$$

Substituting into (1):

$$\frac{dY}{dX} = \frac{(X + h) + 2(Y + k) - 3}{2(X + h) + (Y + k) - 3} = \frac{X + 2Y + (h + 2k - 3)}{2X + Y + (2h + k - 3)}$$

Now choose constants h and k such that:

$$\begin{cases} h + 2k - 3 = 0 \\ 2h + k - 3 = 0 \end{cases} \implies h = 1, \quad k = 1$$

Therefore:

$$\frac{dY}{dX} = \frac{X + 2Y}{2X + Y} \quad [\text{Homogeneous in } X \text{ and } Y]$$

Put $Y = vX \implies \frac{dY}{dX} = v + X \frac{dv}{dX}$:

$$v + X \frac{dv}{dX} = \frac{X + 2vX}{2X + vX} = \frac{1 + 2v}{2 + v} \implies X \frac{dv}{dX} = \frac{1 + 2v}{2 + v} - v = \frac{1 - v^2}{2 + v}$$

$$\implies \frac{dX}{X} = \left(\frac{2 + v}{1 - v^2} \right) dv = \left(\frac{2}{1 - v^2} + \frac{v}{1 - v^2} \right) dv$$

Integrating:

$$\ln X = 2 \cdot \frac{1}{2} \ln \left| \frac{1 + v}{1 - v} \right| - \frac{1}{2} \ln(1 - v^2) + \ln c$$

Reducible to Homogeneous Form (Continued):

Continuing the algebraic simplification:

$$\begin{aligned}\ln X &= \ln \left(\frac{1+v}{1-v} \right) - \ln \sqrt{1-v^2} + \ln c = \ln \left[c \cdot \frac{1+v}{1-v} \cdot \frac{1}{\sqrt{1+v}\sqrt{1-v}} \right] \\ \Rightarrow X &= \frac{c\sqrt{1+v}}{(1-v)^{3/2}} \Rightarrow X^2(1-v)^3 = c^2(1+v) \quad \left[\text{since } v = \frac{Y}{X} \right] \\ \Rightarrow X^2 \left(1 - \frac{Y}{X} \right)^3 &= c^2 \left(1 + \frac{Y}{X} \right) \Rightarrow (X-Y)^3 = c^2(X+Y)\end{aligned}$$

Substituting back $X = x - 1$ and $Y = y - 1$:

$$\begin{aligned}\{(x-1) - (y-1)\}^3 &= c^2\{(x-1) + (y-1)\} \\ \Rightarrow (x-y)^3 &= c^2(x+y-2) \quad \text{which is the required solution.}\end{aligned}$$

Solved Problems and Exercises (S.L. Ross / B.D. Sharma)

First-Order Linear Differential Equations:

A differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x) \tag{13}$$

is called a **linear differential equation of the first order**.

Method of Solution:

To solve (13), multiply both sides by $e^{\int P(x)dx}$:

$$\begin{aligned}e^{\int P(x)dx} \frac{dy}{dx} + P(x)y e^{\int P(x)dx} &= Q(x)e^{\int P(x)dx} \\ \Rightarrow \frac{d}{dx} \left(y e^{\int P(x)dx} \right) &= Q(x)e^{\int P(x)dx}\end{aligned}$$

Integrating both sides with respect to x :

$$y e^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + c$$

which gives the required solution.

Integrating Factor (I.F.):

The factor $e^{\int P(x)dx}$ which transforms the non-integrable left side into an exact derivative is called the **Integrating Factor (I.F.)**.

Linear DE Examples:

Example 1: Solve $x \frac{dy}{dx} + 2y = x^2 \log x$ — (1)

Solution: Standardizing (1):

$$\frac{dy}{dx} + \frac{2}{x}y = x \log x$$

$$\therefore \text{I.F.} = e^{\int \frac{2}{x} dx} = e^{2 \log x} = e^{\log x^2} = x^2$$

Multiplying both sides by x^2 :

$$x^2 \frac{dy}{dx} + 2xy = x^3 \log x \implies \frac{d}{dx}(yx^2) = x^3 \log x$$

Integrating both sides:

$$x^2 y = \int x^3 \log x \, dx + c = c + \log x \cdot \frac{x^4}{4} - \int \frac{1}{x} \cdot \frac{x^4}{4} dx$$

$$\implies x^2 y = c + \frac{x^4}{4} \log x - \frac{1}{4} \cdot \frac{x^4}{4} = c + \frac{x^4}{4} \left(\log x - \frac{1}{4} \right)$$

$$\implies y = cx^{-2} + \frac{1}{4}x^2 \left(\log x - \frac{1}{4} \right) \quad \text{which is the required solution.}$$

Example 2 (IVP): Solve $(x^2 + 1)\frac{dy}{dx} + 4xy = x$, with $y(2) = 1$ — (2)

Solution: Standardizing:

$$\frac{dy}{dx} + \frac{4x}{x^2 + 1}y = \frac{x}{x^2 + 1} \quad \text{--- (3)}$$

$$\text{I.F.} = e^{\int \frac{4x}{x^2 + 1} dx} = e^{2\ln(x^2 + 1)} = e^{\ln(x^2 + 1)^2} = (x^2 + 1)^2$$

Multiplying equation (3) by I.F. = $(x^2 + 1)^2$:

$$(x^2 + 1)^2 \frac{dy}{dx} + \frac{4x}{x^2 + 1} (x^2 + 1)^2 y = \frac{x}{x^2 + 1} (x^2 + 1)^2$$

$$\implies (x^2 + 1)^2 \frac{dy}{dx} + 4x(x^2 + 1)y = x(x^2 + 1)$$

$$\implies \frac{d}{dx} [(x^2 + 1)^2 y] = x^3 + x$$

Integrating both sides:

$$(x^2 + 1)^2 y = \int (x^3 + x) dx = \frac{x^4}{4} + \frac{x^2}{2} + c \quad \text{--- (4)}$$

Applying initial condition $y(2) = 1$:

$$(2^2 + 1)^2 \cdot 1 = \frac{2^4}{4} + \frac{2^2}{2} + c \implies 25 = 4 + 2 + c \implies c = 19$$

Substituting $c = 19$ into (4):

$$(x^2 + 1)^2 y = \frac{x^4}{4} + \frac{x^2}{2} + 19 \quad \text{Ans.}$$

Bernoulli's Differential Equation:

An equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad \text{--- (1)} \tag{14}$$

is called a **Bernoulli Differential Equation**.

Divide (1) by y^n :

$$y^{-n} \frac{dy}{dx} + P(x)y^{-n+1} = Q(x) \quad \text{--- (2)}$$

Put $y^{-n+1} = v \implies (-n+1)y^{-n} \frac{dy}{dx} = \frac{dv}{dx} \implies y^{-n} \frac{dy}{dx} = \frac{1}{1-n} \frac{dv}{dx}$.

Substituting into (2):

$$\frac{1}{1-n} \frac{dv}{dx} + P(x)v = Q(x) \implies \frac{dv}{dx} + (1-n)P(x)v = (1-n)Q(x)$$

which is a linear equation in v and x .

Non-Linear Equations Reducible to Linear Form:

An equation of the form:

$$f'(y) \frac{dy}{dx} + P(x)f(y) = Q(x) \quad \text{--- (1)}$$

Put $f(y) = v \implies f'(y) \frac{dy}{dx} = \frac{dv}{dx}$.

Then (1) becomes: $\frac{dv}{dx} + P(x)v = Q(x)$, which is linear in v and x .

Bernoulli Equation Examples:

Example 1: Solve $\frac{dy}{dx} + xy = xy^3$

Solution: This is a Bernoulli equation with $n = 3$.

Dividing by y^3 (or multiplying by y^{-3}):

$$y^{-3} \frac{dy}{dx} + y^{-2}x = x \quad \text{--- (1)}$$

Let $v = y^{1-n} = y^{1-3} = y^{-2} \implies \frac{dv}{dx} = -2y^{-3} \frac{dy}{dx} \implies y^{-3} \frac{dy}{dx} = -\frac{1}{2} \frac{dv}{dx}$.

Equation (1) becomes:

$$-\frac{1}{2} \frac{dv}{dx} + vx = x \implies \frac{dv}{dx} - 2vx = -2x \quad \text{--- (2)}$$

$$\text{I.F.} = e^{\int -2x dx} = e^{-x^2}$$

Multiplying (2) by e^{-x^2} :

$$e^{-x^2} \frac{dv}{dx} - 2xe^{-x^2}v = -2xe^{-x^2} \implies \frac{d}{dx} (ve^{-x^2}) = -2xe^{-x^2}$$

Integrating both sides:

$$ve^{-x^2} = \int -2xe^{-x^2} dx + c = e^{-x^2} + c$$

$$\implies v = 1 + ce^{x^2}$$

Since $v = \frac{1}{y^2}$, we obtain:

$$\frac{1}{y^2} = 1 + ce^{x^2} \quad \text{Ans.}$$

Bernoulli Equation Examples (Continued):

Example 2: Solve $x \frac{dy}{dx} + y = y^2 \log x$

Solution: Dividing by xy^2 :

$$y^{-2} \frac{dy}{dx} + \frac{1}{x} y^{-1} = \frac{1}{x} \log x \quad \text{--- (1)}$$

Put $-y^{-1} = v \implies \frac{dv}{dx} = y^{-2} \frac{dy}{dx}$.

Substituting into (1):

$$\frac{dv}{dx} - \frac{1}{x} v = \frac{1}{x} \log x \quad \text{--- (2)}$$

$$\text{I.F.} = e^{\int -\frac{1}{x} dx} = e^{-\log x} = \frac{1}{x}$$

Multiplying (2) by $\frac{1}{x}$:

$$\frac{1}{x} \frac{dv}{dx} - \frac{1}{x^2} v = \frac{1}{x^2} \log x \implies \frac{d}{dx} \left(v \cdot \frac{1}{x} \right) = \frac{1}{x^2} \log x$$

Integrating both sides:

$$v \cdot \frac{1}{x} = c + \int \frac{1}{x^2} \log x \, dx = c + \log x \left(-\frac{1}{x} \right) - \int \frac{1}{x} \left(-\frac{1}{x} \right) dx$$

$$\implies \frac{v}{x} = c - \frac{1}{x} \log x - \frac{1}{x} \implies -v = (1 + \log x) - cx$$

Since $v = -y^{-1}$:

$$\frac{1}{y} = (1 + \log x) - cx \quad \text{Ans.}$$

Exact Differential Equations:

Total Differential:

If the function $F(x, y)$ has continuous first partial derivatives in a domain D , then the **total differential** dF is defined by:

$$dF(x, y) = \frac{\partial F(x, y)}{\partial x} dx + \frac{\partial F(x, y)}{\partial y} dy \quad \forall (x, y) \in D$$

Exact Differential:

The expression $M(x, y)dx + N(x, y)dy$ is called an **exact differential** if there exists a function $F(x, y)$ such that:

$$dF(x, y) = M(x, y)dx + N(x, y)dy$$

Exact Differential Equations & Trajectories

Lecture Notes — Section 2: Exactness, Integrating Factors & Higher Order LODEs

Exact Differential Equations:

If the total differential of a function $F(x, y)$ is given by:

$$dF(x, y) = M(x, y)dx + N(x, y)dy \quad \forall (x, y) \in D$$

then $M(x, y) = \frac{\partial F}{\partial x}$ and $N(x, y) = \frac{\partial F}{\partial y}$.

Definition (Exact DE): If $M(x, y)dx + N(x, y)dy$ is an exact differential, then the differential equation

$$M(x, y)dx + N(x, y)dy = 0 \quad \text{--- (15)}$$

is called an **exact differential equation**.

Theorem (Necessary and Sufficient Condition):

Consider the DE $M(x, y)dx + N(x, y)dy = 0$, where $M(x, y)$ and $N(x, y)$ have continuous first partial derivatives $\forall (x, y) \in D$.

If the DE (15) is exact in D , then:

$$\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x} \quad \forall (x, y) \in D$$

Conversely, if $\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x} \quad \forall (x, y) \in D$, then the DE (15) is exact in D .

Working Rule for Solving Exact Differential Equations:

If the DE $M(x, y)dx + N(x, y)dy = 0$ satisfies the condition $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, it is exact. To integrate it:

- (i) Integrate $M(x, y)$ with respect to x , regarding y as constant.
- (ii) Find out those terms in $N(x, y)$ which are free from x and integrate them with respect to y .
- (iii) Add the two expressions so obtained and equate the sum to an arbitrary constant c .

This gives the general solution of the given exact DE.

Example 1: Solve $(3x^2 + 4xy)dx + (2x^2 + 2y)dy = 0$

Solution: Here, $M(x, y) = 3x^2 + 4xy$ and $N(x, y) = 2x^2 + 2y$.

$$\frac{\partial M}{\partial y} = 4x \quad \text{and} \quad \frac{\partial N}{\partial x} = 4x \implies \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 4x \quad \forall (x, y)$$

Thus, the given equation is exact in every rectangular domain D .

1. Integrating M w.r.t. x regarding y as constant:

$$\int M(x, y)dx = \int (3x^2 + 4xy)dx = x^3 + 2x^2y$$

2. The term in $N(x, y)$ free from x is $2y$. Integrating this w.r.t. y :

$$\int 2y dy = y^2$$

Adding these two and equating to an arbitrary constant:

$$x^3 + 2x^2y + y^2 = c \quad \text{Ans.}$$

Note: You may also refer to the solution methodology presented in S.L. Ross.

Example 2 (IVP): Solve the IVP:

$$(2x \cos y + 3x^2y)dx + (x^3 - x^2 \sin y - y)dy = 0, \quad \text{with } y(0) = 2$$

Solution:

$$\frac{\partial M}{\partial y} = -2x \sin y + 3x^2 = \frac{\partial N}{\partial x} \implies \text{The DE is exact.}$$

$$1. \int (2x \cos y + 3x^2y)dx = x^2 \cos y + x^3y$$

$$2. \int (-y)dy = -\frac{y^2}{2}$$

Adding the results:

$$x^2 \cos y + x^3y - \frac{y^2}{2} = c$$

Applying the initial condition $x = 0, y = 2$:

$$0^2 \cos(2) + 0^3(2) - \frac{2^2}{2} = c \implies c = -2$$

Substituting $c = -2$:

$$x^2 \cos y + x^3y - \frac{y^2}{2} = -2 \quad \text{Ans.}$$

Integrating Factor (I.F.):

If the differential equation $M(x, y)dx + N(x, y)dy = 0$ — (1) is not exact, but becomes exact when multiplied by a factor $\mu(x, y)$:

$$\mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy = 0 \quad \text{— (2)} \quad (16)$$

then $\mu(x, y)$ is called an **Integrating Factor (I.F.)** of (1).

Rules for Finding Integrating Factors:

Rule 1: If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$ (a function of x only), then I.F. = $e^{\int f(x)dx}$.

Rule 2: If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M} = g(y)$ (a function of y only), then I.F. = $e^{\int g(y)dy}$.

Rule 3: If $Mdx + Ndy = 0$ is homogeneous and $Mx + Ny \neq 0$, then I.F. = $\frac{1}{Mx+Ny}$.

Example 3: Solve $(x^2 + y^2 + 1)dx - 2xy dy = 0$ — (1)

Solution: $M = x^2 + y^2 + 1$ and $N = -2xy$.

$$\frac{\partial M}{\partial y} = 2y, \quad \frac{\partial N}{\partial x} = -2y \implies \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \quad (\text{Not Exact})$$

Using Rule 1:

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y - (-2y)}{-2xy} = \frac{4y}{-2xy} = -\frac{2}{x} \quad (\text{Function of } x \text{ alone})$$

$$\therefore \text{I.F.} = e^{\int -\frac{2}{x}dx} = e^{-2\ln x} = \frac{1}{x^2}$$

Multiplying (1) by $\frac{1}{x^2}$:

$$\left(1 + \frac{y^2}{x^2} + \frac{1}{x^2}\right)dx - \frac{2y}{x}dy = 0 \quad \text{— (2)}$$

Now, $\frac{\partial M_2}{\partial y} = \frac{2y}{x^2}$ and $\frac{\partial N_2}{\partial x} = \frac{2y}{x^2}$, so equation (2) is exact.

Integrating $1 + \frac{y^2}{x^2} + \frac{1}{x^2}$ w.r.t. x (keeping y constant):

$$\int \left(1 + \frac{y^2}{x^2} + \frac{1}{x^2}\right)dx = x - \frac{y^2}{x} - \frac{1}{x}$$

There are no terms in $N_2 = -\frac{2y}{x}$ free from x . Hence, the solution is:

$$x - \frac{y^2}{x} - \frac{1}{x} = c \quad \text{Ans.}$$

Example 4: Solve $(x^2 + y^2 + 2x)dx + 2y dy = 0$ — (1)

Solution: $\frac{\partial M}{\partial y} = 2y$ and $\frac{\partial N}{\partial x} = 0$.

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y - 0}{2y} = 1 \implies \text{I.F.} = e^{\int 1 dx} = e^x$$

Multiplying (1) by e^x :

$$e^x(x^2 + y^2 + 2x)dx + e^x \cdot 2y dy = 0 \quad (\text{which is exact})$$

Rewriting the exact equation:

$$\begin{aligned} e^x(x^2 + 2x)dx + y^2 e^x dx + 2y e^x dy &= 0 \\ \implies d(x^2 e^x) + d(y^2 e^x) &= 0 \end{aligned}$$

Integrating both sides:

$$x^2 e^x + y^2 e^x = c \implies e^x(x^2 + y^2) = c \quad \text{Ans.}$$

Trajectories

Let $F(x, y, c) = 0$ — (1) be a given one-parameter family of curves.

- **Oblique Trajectory:** A curve that intersects the curves of the family (1) at a constant angle $\alpha \neq 90^\circ$ is called an oblique trajectory.
- **Orthogonal Trajectory:** A curve that intersects the curves of the family at right angles ($\alpha = 90^\circ$) is called an orthogonal trajectory.

Equation of Trajectories:

If a family of curves is given by the differential equation $f(x, y, p) = 0$ (where $p = \frac{dy}{dx}$), then its trajectory at angle α is given by replacing p with $\frac{p - \tan \alpha}{1 + p \tan \alpha}$:

$$f\left(x, y, \frac{p - \tan \alpha}{1 + p \tan \alpha}\right) = 0$$

Orthogonal Trajectories ($\alpha = 90^\circ$):

For $\alpha = 90^\circ$, we have:

$$\frac{p - \tan \alpha}{1 + p \tan \alpha} = \frac{p \cot \alpha - 1}{\cot \alpha + p} = \frac{-1}{p} \quad [\text{since } \cot 90^\circ = 0]$$

Orthogonal Trajectories (Continued):

Hence, corresponding to the family of curves $f(x, y, p) = 0$, the differential equation of the orthogonal trajectories is:

$$f\left(x, y, -\frac{1}{p}\right) = 0 \quad \text{or} \quad f\left(x, y, -\frac{dx}{dy}\right) = 0$$

Integration of this equation yields the family of orthogonal trajectories.

Polar Coordinates:

If a family of curves is given in polar form by $f\left(r, \theta, \frac{dr}{d\theta}\right) = 0$, then the differential equation of its orthogonal trajectories is given by replacing $\frac{dr}{d\theta}$ with $-r^2 \frac{d\theta}{dr}$:

$$f\left(r, \theta, -r^2 \frac{d\theta}{dr}\right) = 0$$

Example 1: Find the orthogonal trajectories of the family of circles $x^2 + y^2 = c^2$ — (1)

Solution: Differentiating (1) with respect to x :

$$2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y} \quad \text{— (2) [DE of given family]}$$

Replacing $\frac{dy}{dx}$ by its negative reciprocal $-\frac{dx}{dy}$ (i.e., replacing $-\frac{x}{y}$ with $\frac{y}{x}$):

$$\frac{dy}{dx} = \frac{y}{x} \quad \text{— (3) [DE of orthogonal trajectories]}$$

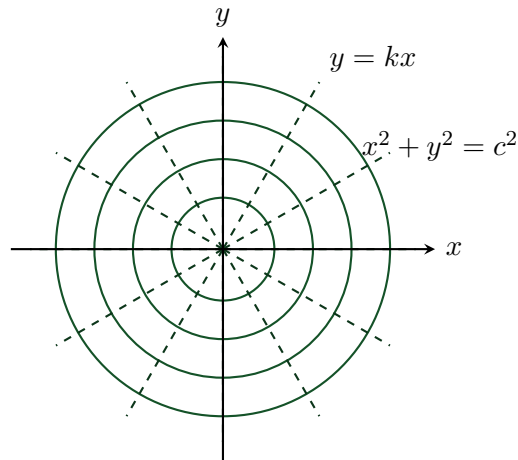
Separating variables in (3):

$$\frac{dy}{y} = \frac{dx}{x}$$

Integrating both sides:

$$\ln y = \ln x + \ln k \implies y = kx \quad \text{Ans.}$$

This single-parameter family $y = kx$ represents straight lines passing through the origin, which are orthogonal to the family of concentric circles $x^2 + y^2 = c^2$.



Example 2: Find the orthogonal trajectories of $y = cx^2$.

Solution: Differentiating $y = cx^2 \implies \frac{dy}{dx} = 2cx$. Since $c = \frac{y}{x^2}$, we get $\frac{dy}{dx} = 2 \left(\frac{y}{x^2} \right) x = \frac{2y}{x}$. Replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$:

$$\frac{dy}{dx} = -\frac{x}{2y} \implies 2y \, dy + x \, dx = 0$$

Integrating:

$$y^2 + \frac{x^2}{2} = k_0 \implies x^2 + 2y^2 = k \quad \text{Ans.}$$

Exercise (S.L. Ross P.75): Find a family of oblique trajectories of the family of straight lines $y = cx$ at an angle $\alpha = 45^\circ$.

Higher Order Linear Differential Equations:

A linear ordinary differential equation (LODE) of order n in the dependent variable y and independent variable x is an equation that can be expressed as:

$$a_0(x)\frac{d^n y}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_{n-1}(x)\frac{dy}{dx} + a_n(x)y = F(x) \quad (1) \quad (17)$$

where $a_0(x) \neq 0$.

We assume that $a_0, a_1, \dots, a_n$ are continuous on an interval $a \leq x \leq b$ and $a_0(x) \neq 0$ for all $x \in [a, b]$. The function $F(x)$ is called the **non-homogeneous term**.

If $F(x) = 0$, then equation (17) reduces to:

$$a_0(x)\frac{d^n y}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_{n-1}(x)\frac{dy}{dx} + a_n(x)y = 0 \quad (2) \quad (18)$$

and is called a **homogeneous linear differential equation**.

Existence and Uniqueness Theorem

Let x_0 be any point on $a \leq x \leq b$ and let $c_0, c_1, \dots, c_{n-1}$ be n arbitrary constants. Then there exists a **unique solution** $f(x)$ of equation (17) satisfying the initial conditions:

$$f(x_0) = c_0, \quad f'(x_0) = c_1, \quad \dots, \quad f^{(n-1)}(x_0) = c_{n-1}$$

which is defined throughout $a \leq x \leq b$.

Note: If f is a solution of the homogeneous equation (18) with $f(x_0) = 0, f'(x_0) = 0, \dots, f^{(n-1)}(x_0) = 0$, then $f(x) = 0$ identically on $a \leq x \leq b$.

Linear Combination and General Solution:

Definition (Linear Combination): If $f_1, f_2, \dots, f_m$ are m functions and $c_1, c_2, \dots, c_m$ are m constants, then the expression:

$$c_1 f_1 + c_2 f_2 + \cdots + c_m f_m$$

is called a **linear combination** of $f_1, f_2, \dots, f_m$.

Property: If $f_1, f_2, \dots, f_m$ are m solutions of homogeneous DE (18), then any linear combination $c_1 f_1 + c_2 f_2 + \cdots + c_m f_m$ is also a solution of (18).

Fundamental Theorem of Homogeneous LODEs

The n^{th} order homogeneous linear DE (18) always possesses n linearly independent solutions.

Furthermore, if $f_1, f_2, \dots, f_n$ are n linearly independent solutions of (18), then **every solution** of (18) can be expressed as a linear combination:

$$c_1 f_1 + c_2 f_2 + \cdots + c_n f_n$$

by a proper choice of constants $c_1, c_2, \dots, c_n$.

General Solution: If $f_1, f_2, \dots, f_n$ are n linearly independent solutions of (18) on $a \leq x \leq b$, then:

$$f(x) = c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x)$$

(where $c_1, c_2, \dots, c_n$ are arbitrary constants) is called the **general solution** of (18).

The Wronskian

Let $f_1, f_2, \dots, f_n$ be n real functions, each having at least $n - 1$ derivatives on $a \leq x \leq b$. The determinant:

$$W(f_1, f_2, \dots, f_n) = \begin{vmatrix} f_1 & f_2 & \dots & f_n \\ f_1' & f_2' & \dots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \dots & f_n^{(n-1)} \end{vmatrix}$$

is called the **Wronskian** of the functions $f_1, f_2, \dots, f_n$.

If the Wronskian $W(f_1, f_2, \dots, f_n) \neq 0$ for at least one point on $a \leq x \leq b$, then the functions are **linearly independent**.

Example: Show that the solutions e^x , e^{-x} , and e^{2x} of the DE

$$\frac{d^3 y}{dx^3} - 2 \frac{d^2 y}{dx^2} - \frac{dy}{dx} + 2y = 0$$

are linearly independent.

Solution: Evaluating the Wronskian of e^x, e^{-x}, e^{2x} :

$$W(e^x, e^{-x}, e^{2x}) = \begin{vmatrix} e^x & e^{-x} & e^{2x} \\ e^x & -e^{-x} & 2e^{2x} \\ e^x & e^{-x} & 4e^{2x} \end{vmatrix}$$

Factoring out e^x, e^{-x}, e^{2x} from columns 1, 2, and 3:

$$W = (e^x)(e^{-x})(e^{2x}) \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 1 & 1 & 4 \end{vmatrix}$$

$$= e^{2x} [1(-4 - 2) - 1(4 - 2) + 1(1 + 1)] = e^{2x} (-6 - 2 + 2) = -6e^{2x} \neq 0 \quad \forall x \in \mathbb{R}$$

Thus, since $W(e^x, e^{-x}, e^{2x}) \neq 0$ for all real values of x , e^x, e^{-x}, e^{2x} are linearly independent solutions of the given differential equation on every real interval.